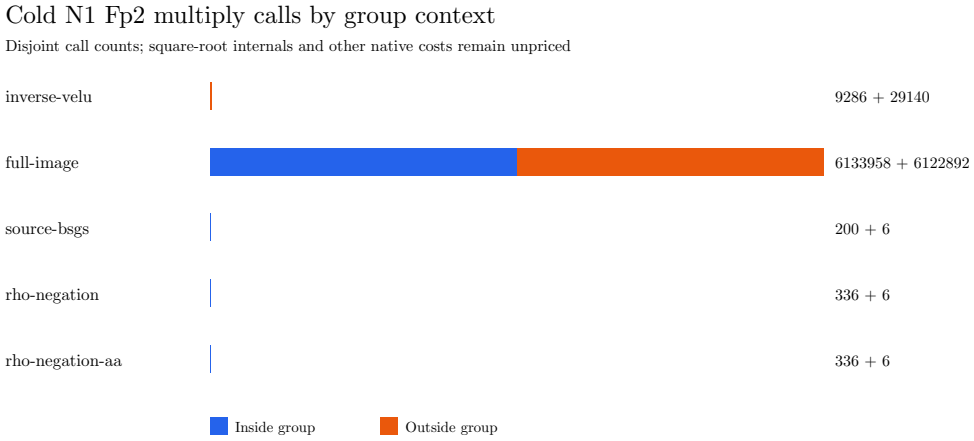
Disjoint fiber price and Linux replay

2026-10-16 · Native Rust · source-linked report

The instrumented order-1,201 panel completed 60/60 cold children. A separate Rust checker replayed 1,275 scalars and 6,126 relation witnesses; the native audit found 60/60 deterministic matches with the frozen V2 answers, rows, group counts and earlier field totals. The public input, protocol, raw children, audit and source snapshot are beside this report under `research/isogeny_torsion_fibers_20261012/primitive_price_20261016/`. The audit's first failed reducer attempt is retained there as V1; its corrected result is `PRIMITIVE_AUDIT_V2.json`.

The generic two-class floor is $S = \sqrt{\frac{\pi}{4}} = 0.886227$ in counted group additions per $\sqrt{r}$. Same-target signed rho and source BSGS are the references, with rho A/A as a noise control. The graph displays the top-level F_{p^2} multiplication calls split by whether they occurred inside a curve addition or doubling. It does not convert these calls, square roots, allocations, or process work into one portable unit.



Targets	Arm	S_{group} mean	Inside F_{p^2} mul	Outside F_{p^2} mul
1	Full image	88,499.672	6,133,958	6,122,892
1	Inverse Vélu	134.457	9,286	29,140
1	Source BSGS	2.972	200	6
1	Signed rho	4.905	336	6
64	Full image	98,067.547	6,796,835	6,123,144
64	Inverse Vélu	9,702.332	672,163	29,392
64	Source BSGS	84.171	5,696	132
64	Signed rho	259.728	17,853	132

At one target, the full-image/inverse counted-group ratio is 658.201 and inverse/rho is 27.410 using three-round means. These are group count diagnostics. The group counts overlap the field call ledger and must not be added to it. The inside/outside split places 29,140 Fp² multiply calls in inverse setup and polynomial work versus six in rho; the internal `sqrt_mod`, polynomial control flow, table work, allocation, I/O and cross-host calibration are still open. Portable complete S remains unset.

Linux independent-host receipts

GitHub Actions run 38961930762 replayed all 60 order-1,201 children, 1,275 scalars and 6,126 witnesses. Run 38961930725 replayed all 108 order-229 nine-arm rows, 996 witnesses and 2,484 trials. The artifacts and SHA-256 manifests are copied into `LINUX_RECEIPTS_V1/`. Their Ubuntu x86-64 AMD EPYC virtual host supplies independent L1 correctness and descriptive performance; no physical-core L2 wall admission exists.

Run 38961930860 executed the hard-capped order-2,437 full-image probe. It exited 134 after a 1,072-byte allocation failed, with peak RSS 534,470,656 bytes and no solver stdout. This is a resource-censored row under the 512 MiB address-space cap. A memory-bounded constructor is needed for the next same-cap attempt.

The next DLP experiment will freeze a projected destination factor base without a raw preimage witness and compare its complete verified solve with source rho, BSGS and relation controls. A changed base can change relation support; its total cost must be measured directly.